Practice workbook · Sketching curves and where they cross the axes

Key points and freehand sketches of non-rational curves: parabolas, cubics, exponential and log curves.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-GRA-03

1. Question 3 · part e · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part e only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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1. Question 3 · part e · 2 marks

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Worked-example retry · Attempt part e only.

Question 3 continued

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1. Question 3 · part e · 2 marks

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Worked-example retry · Attempt part e only.

Question 3 continued

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1. Question 3 · part e · 2 marks

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Worked-example retry · Attempt part e only.



1. Question 3 · part e · 2 marks · Working and answer

Working and answer

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2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

2 (a) Express $x^2 + 4x - 8$ in the form $(x + a)^2 + b$ where a and b are constants whose values are to be found.

(2)

(b) Use algebra to solve the simultaneous equations

$$y = x^2 + 4x - 8$$
$$y = 2x + 7$$

(5)

The curve C has equation $y = x^2 + 4x - 8$
The straight line L has equation $y = 2x + 7$

Using the same axes and the results of part (a) and part (b),

(c) sketch C and L, showing clearly the coordinates of the turning point of C and the coordinates of the points of intersection of C and L.

(4)

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2. Question 2 · part c · 4 marks

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Further practice · Attempt part c only.

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Question 2 continued

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2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 2 continued

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8

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2. Question 2 · part c · 4 marks

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Further practice · Attempt part c only.



Question 2 continued

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(Total for Question 2 is 11 marks)

2. Question 2 · part c · 4 marks · Working and answer

Working and answer

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3. Question 11 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

11

Diagram **NOT** accurately drawn

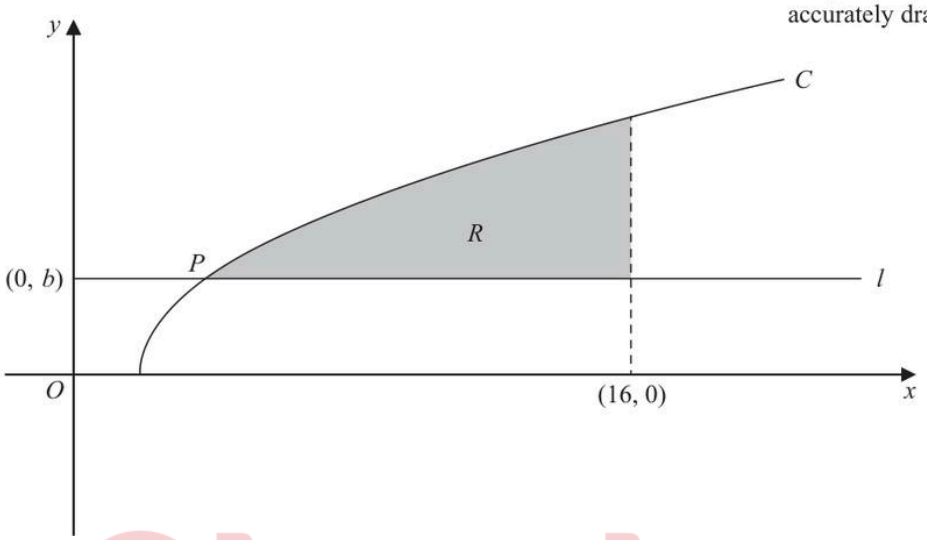


Figure 2

Figure 2 shows part of the curve C with equation $y = \sqrt{x - 2}$
Figure 2 also shows the straight line l with equation $y = b$ for $x > 0$ where $b > 0$

Given that C and l intersect at the point P with coordinates (a, b), where $2 < a < 16$

(a) show that $b^2 = a - 2$

(2)

The finite region R bounded by C, the straight line with equation $x = 16$ and l, shown shaded in Figure 2, is rotated through 360° about the x-axis to form a solid S.

Given that the volume of the solid formed is 50π

(b) use algebraic integration to find the value of a and the value of b.

(9)

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3. Question 11 · part a · 2 marks · Working and answer

Working and answer

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4. Question 11 · part a · 1 mark

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

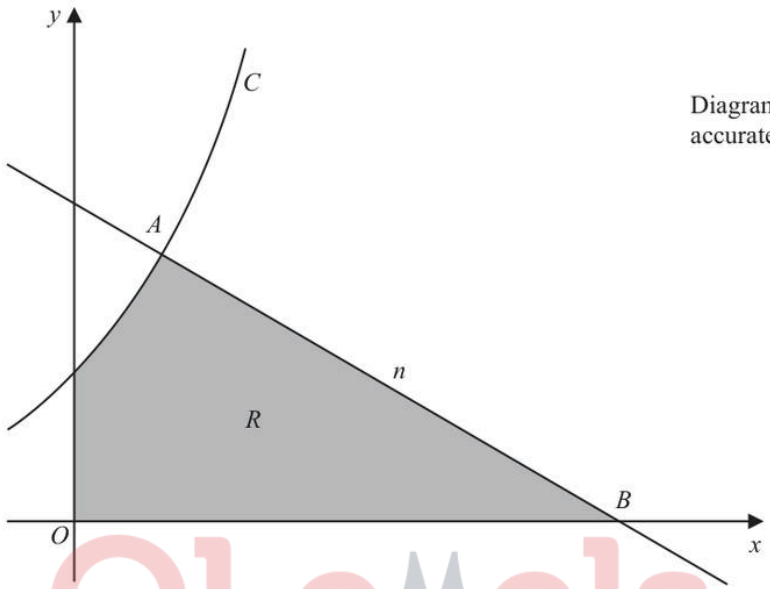


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a curve C with equation $y = 3e^{\frac{x}{3}}$
The point A with coordinates $(2, p)$ lies on C

(a) Write down the exact value of p

(1)

The straight line n, shown on Figure 4, is the normal to C at A and crosses the x-axis at point B

The region R, shown shaded in Figure 4, is bounded by C, n, the x-axis and the y-axis.

(b) Use algebraic integration to find the area of R

Give your answer in the form $\frac{W}{V}e^2 + We^{\frac{2}{3}} - W$

where W and V are integers to be found.

(11)

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4. Question 11 · part a · 1 mark · Working and answer

Working and answer

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5. Question 11 · part a.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part a.i only.

11

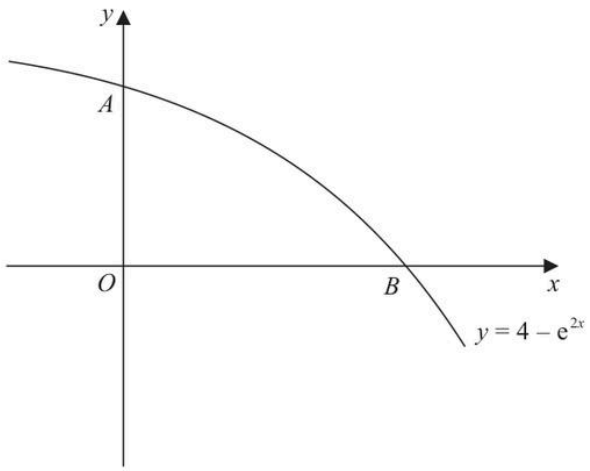


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Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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5. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 11 · part a.i · 0 marks

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Further practice · Attempt part a.i only.

11

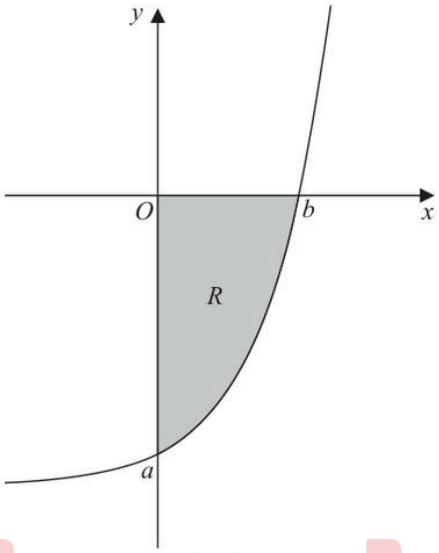


Diagram **NOT** accurately drawn

Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where p and q are integers.

(6)

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6. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

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7. Question 7 · part a · 4 marks

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Further practice · Attempt part a only.

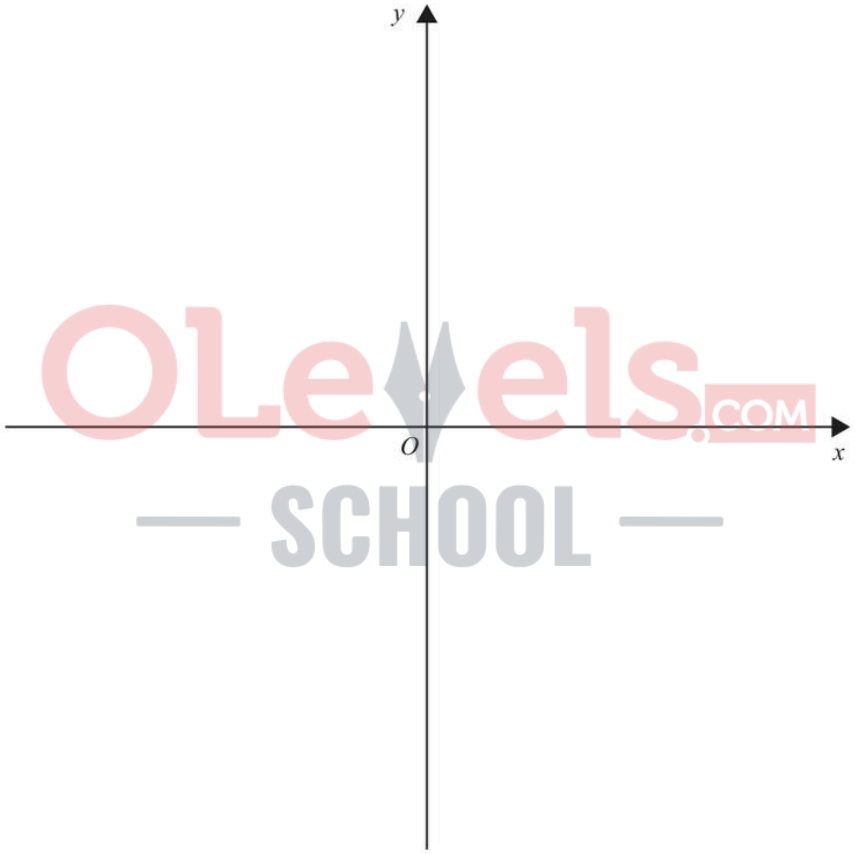
7 The curve C has equation $y = -\log_4(x + 4)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes and the equation of any asymptote to C.

(4)

(b) Solve the equation $\log_{(x+4)} 256 - \log_4(x + 4) = 0$

(5)



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7. Question 7 · part a · 4 marks · Working and answer

Working and answer

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8. Question 7 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 A curve C has equation $y = \log_{10}(x + 2)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes.

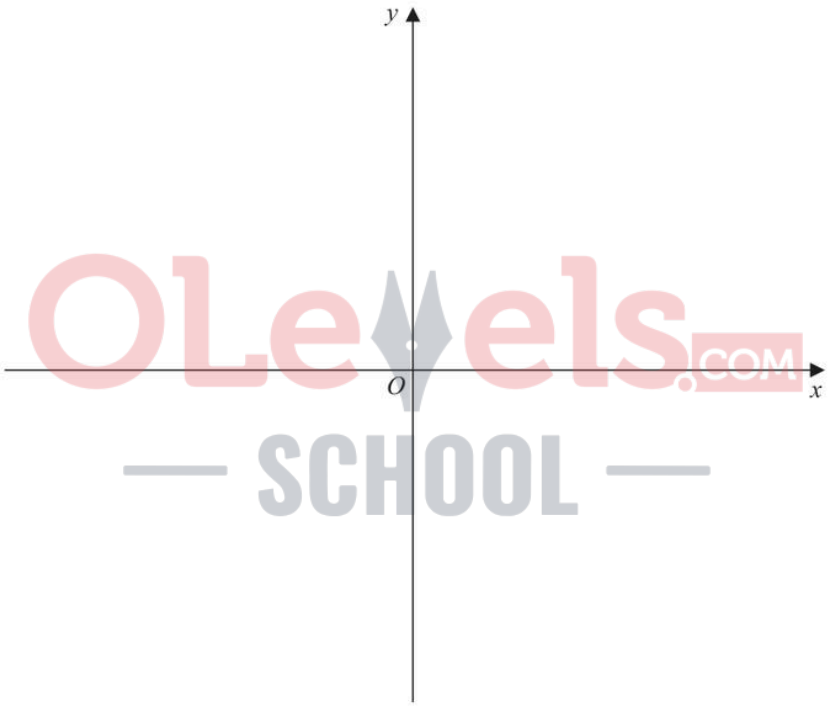
(2)

(b) Solve the equation $2(\log_a 4 + \log_a 16) = 1$

(3)

(c) Solve the equation $5\log_q 16 + 4\log_2 q = 24$

(6)



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8. Question 7 · part a · 2 marks · Working and answer

Working and answer

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9. Question 6 · part a · 3 marks

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Further practice · Attempt part a only.

6

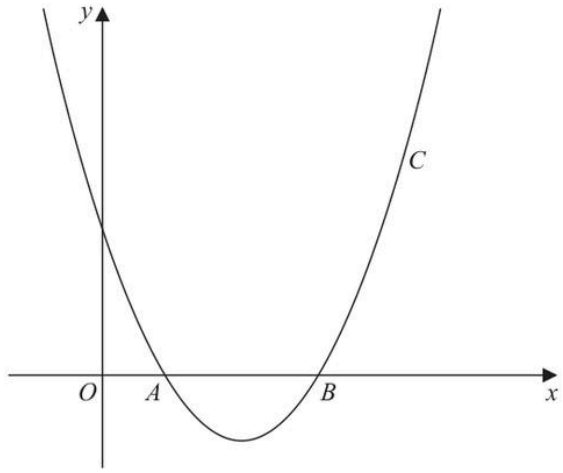


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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9. Question 6 · part a · 3 marks · Working and answer

Working and answer

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10. Question 3 · part c · 2 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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10. Question 3 · part c · 2 marks · Working and answer

Working and answer

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11. Question 5 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

5

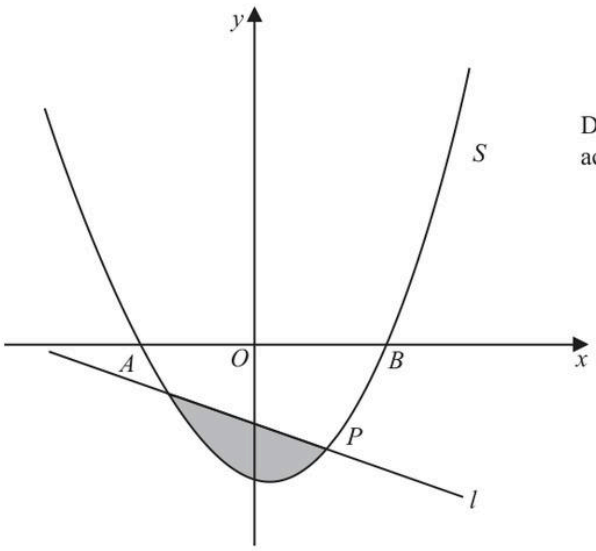


Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

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11. Question 5 · part a · 3 marks · Working and answer

Working and answer

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